

CSCE 647 IMAGE SYNTHESIS · MODULE 07 · LECTURE NOTES

BSDFs and Microfacet Theory

Where materials come from, and how to sample them.

MODULE OUTCOMES

Derive the microfacet BRDF and explain each of its three terms.

Explain why GGX displaced Beckmann.

Implement visible-normal sampling and its PDF.

Handle specular surfaces correctly in a Monte Carlo renderer.

Verify a BSDF for energy conservation and reciprocity.

1 Where the model comes from

A ROUGH SURFACE IS A FIELD OF TINY MIRRORS

The microfacet assumption is that a surface which appears rough is, at a scale below what a pixel can resolve, composed of many perfectly specular facets with varying orientations.

You never see an individual facet. You see the **statistical aggregate** of their reflections, which is why a rough metal has a broad highlight and a polished one has a sharp reflection — the same physics, different orientation statistics.

So '**roughness**' is a **distribution of facet normals**, not a blur radius chosen to taste. This is the reason microfacet models look convincing where earlier ad-hoc models looked like plastic: the shape of the highlight, the brightening at grazing angles, and the relationship between roughness and highlight size all *follow* from the geometry rather than being independently tuned.

It also explains why they generalise. A model derived from an assumption makes predictions in configurations it was never fitted to.

$$f(\omega_i, \omega_o) = D(h) \cdot F(\omega_o \cdot h) \cdot G(\omega_i, \omega_o) / (4 |\mathbf{n} \cdot \omega_i| |\mathbf{n} \cdot \omega_o|)$$

where $h = \text{normalize}(\omega_i + \omega_o)$ is the **half vector**. The derivation rests on one observation: **only a facet whose normal is exactly h can reflect ω_i into ω_o** . Everything else is counting how many such facets there are, how much they reflect, and how many of them are actually visible.

Term	Answers	Without it
D(h) — normal distribution	What fraction of facets are oriented along h ?	No concept of roughness. This term is the material's appearance.

Term	Answers	Without it
$F(\omega_o \cdot \mathbf{h})$ — Fresnel	What fraction of light striking such a facet is reflected rather than transmitted or absorbed?	No grazing-angle brightening. Every surface becomes mirror-like at glancing incidence in reality, and this term is why.
G — masking-shadowing	What fraction of those facets is neither hidden from the light nor hidden from the viewer by neighbouring facets?	Energy is created at grazing angles. The furnace test fails loudly, with values well above 1.
$4 \mathbf{n} \cdot \omega_i \mathbf{n} \cdot \omega_o $	Normalisation from the change of variables between half-vector space and direction space.	A constant factor error that varies with angle — the worst kind, because it cannot be scaled away.

1.1 Why GGX displaced Beckmann

	Beckmann	GGX / Trowbridge-Reitz
Origin	Derived from Gaussian surface-height statistics; physically motivated by surface metrology.	Originally an empirical fit (Trowbridge-Reitz 1975), reintroduced to graphics by Walter et al. in 2007.
Tail behaviour	Falls off quickly away from the peak.	Heavy tails — significant energy far from the specular direction.
Against measured data	Highlights terminate too abruptly; measured materials have more energy in the tail than it predicts.	Matches measurements substantially better , particularly the soft glow surrounding a highlight.
Status	Historical.	Universal since about 2012. Every production renderer and every real-time engine.

This is a case worth noticing: the empirically fitted distribution beat the physically derived one, because the physical derivation's assumption — Gaussian height statistics — does not describe real machined and worn surfaces as well as the fit does. Deriving from first principles does not guarantee a better model than measuring.

$$D(\mathbf{h}) = \alpha^2 / (\pi ((\mathbf{n} \cdot \mathbf{h})^2 (\alpha^2 - 1) + 1)^2)$$

By convention $\alpha = \text{roughness}^2$, where roughness is the parameter exposed to artists. The squaring makes the control perceptually linear — equal steps in the slider produce roughly equal perceived changes, which they do not if α is exposed directly.

$$\Lambda(\omega) = (-1 + \sqrt{1 + \alpha^2 \tan^2 \theta}) / 2 \quad G_2 = 1 / (1 + \Lambda(\omega_i) + \Lambda(\omega_o))$$

This is the **height-correlated** Smith masking-shadowing function. The older separable form $G_1(\omega_i) \cdot G_1(\omega_o)$ assumes masking and shadowing are independent, which they are not — a facet visible to the viewer is more likely to be visible to the light. The correlated form is more accurate and costs nothing additional. Many implementations still use the separable version out of habit.

THE SINGLE-SCATTERING MODEL LOSES A GREAT DEAL OF ENERGY

The standard microfacet BRDF accounts for light that strikes **one** facet and leaves. In reality, at high roughness, light frequently bounces between several facets before escaping the surface.

That energy is simply discarded. For a rough white metal the loss can exceed 50% — the material renders far darker than it should, and because the loss grows with roughness, the roughness slider doubles as an unwanted brightness control.

The fix in production is Kulla–Conty energy compensation: precompute the directional albedo of the single-scattering model as a function of roughness and viewing angle, store it in a small table, and add back the missing energy as an additional lobe. It is a few hundred lines and is now standard.

The furnace test at high roughness exposes this immediately, which is why Module 05 insisted on running it at several roughness values rather than just one. A model that passes at roughness 0.1 and returns 0.45 at roughness 0.9 has exactly this problem.

2 Sampling a BSDF

Evaluating a BSDF is straightforward. Sampling it well — generating directions distributed approximately proportionally to it, and reporting the correct density — is where the engineering is.

SAMPLE THE VISIBLE NORMALS, NOT ALL OF THEM

The obvious approach samples the distribution $D(h)$ directly, choosing a facet normal and reflecting ω_o about it.

The problem is that many of those facets are not visible from ω_o — they face away, or are masked by neighbouring facets. Samples that select them are wasted, and the proportion wasted grows sharply at grazing angles, which is exactly where microfacet surfaces are most interesting.

Visible-normal sampling (Heitz, 2018) instead draws from the distribution of normals *visible from the outgoing direction*. Every sample selects a facet that can actually contribute.

The result is typically two to three times less variance at grazing angles, for perhaps fifteen extra lines of code. It is strictly better than the older method in every configuration, and there is no longer a reason to implement sampling of D directly.

The PDF of the sampled *direction* — not of the half vector — is:

$$\text{pdf}(\omega_i) = D_{\text{visible}}(h) / (4 |\omega_o \cdot h|)$$

The factor $1/(4|\omega_o \cdot h|)$ is the Jacobian of the reflection operator — the change of variables from the half-vector to the reflected direction. **It is the most commonly omitted term in a BSDF implementation.** Omitting it produces a plausible-looking, converged, wrong image, and no amount of staring at the render will reveal it. The χ^2 test from Module 05 catches it on the first run.

SPECULAR SURFACES REQUIRE A SEPARATE CODE PATH

A perfectly smooth mirror has a BRDF that is a Dirac delta: infinite in the single reflected direction, zero everywhere else.

It cannot be evaluated — $f(\omega_o, \omega_i)$ must return zero, because the probability that an independently chosen ω_i is exactly the mirror direction is zero. **It cannot be importance sampled** with a finite density, because no finite density can represent a delta.

Handle it explicitly: `sample()` returns the reflected direction deterministically, the value $F(\omega_o \cdot n)$, and a PDF of exactly 1 — representing a discrete choice rather than a density.

And set a flag. Next-event estimation (Module 08) must skip specular vertices entirely, because the shadow ray's direction has zero probability of being the mirror direction. MIS must treat the specular strategy as having infinite density so its weight is 1 and the other strategy's is 0. **Getting this wrong double-counts energy**, and the symptom — mirrors that are too bright — is easy to misdiagnose as a Fresnel problem.

Method	Returns	Requirements
$f(\omega_o, \omega_i)$	The BRDF value for a given pair of directions.	Must return zero for specular lobes.
sample(ω_o, u)	A sampled direction, the BRDF value, the PDF, and the lobe flags.	One call, because the three outputs are coupled and recomputing them separately is both slower and a source of inconsistency.
pdf(ω_o, ω_i)	The density this BSDF would have assigned to a direction.	Must be queryable independently of sample(). MIS needs the density of a direction that the <i>light</i> sampler chose.
flags	specular / diffuse / glossy; reflection / transmission.	Drives integrator decisions: whether to do NEE, how to weight, whether the path is still 'specular' for caustic handling.

The independently queryable pdf() is the design requirement most often missed, and omitting it forces a rewrite in Module 08. Design for it now.

3 Beyond reflection

Extension	What it adds	Difficulty
Transmission (BTDF)	Snell's law, total internal reflection, and the Fresnel split between reflected and transmitted energy.	The Jacobian differs from the reflection case, and the BTDF is not symmetric when the refractive index changes — radiance is compressed or expanded by $(\eta_o/\eta_i)^2$. A classic source of bugs in bidirectional renderers.
Rough transmission	The same microfacet machinery with a refraction half vector.	Frosted glass. Walter et al. 2007 gives the complete derivation.
Layered materials	A thin coating over a base: car paint, varnished wood, skin.	Correct treatment requires simulating transport between the layers and is expensive. Most renderers approximate with a weighted sum of lobes, which loses energy and inter-reflection.
Anisotropy	Independent roughness along the two tangent directions.	Brushed metal, hair, satin. Requires a consistent tangent frame from the intersection (Module 03) — and a discontinuous tangent field produces visible seams.
Measured BRDFs	Tabulated data from a gonireflectometer (MERL, RGL databases).	Maximally accurate and awkward: large to store, and importance sampling requires building a numerical CDF over the tabulated data.

4 Testing a BSDF

THREE TESTS, ALL OF THEM NECESSARY

χ^2 test. Generate a million samples with `sample()`, histogram them over the sphere, and compare against the density reported by `pdf()`. This catches any mismatch between the sampler and its stated density — including the missing Jacobian above. **Such a mismatch is completely invisible in a rendered image**, which converges confidently to a wrong answer.

White furnace test. With albedo 1 and no absorption, integrate $f \cdot \cos\theta$ over the hemisphere. It must not exceed 1 (or energy is created) and should be close to 1 for a lossless material. **Run it across the full roughness range** — that is where the single-scattering energy loss of §1 appears, and a single test at low roughness will miss it entirely.

Reciprocity. Evaluate f with the arguments exchanged over a thousand random direction pairs and confirm agreement to floating-point tolerance. Required for the bidirectional methods of Module 11.

A BSDF passing all three is almost certainly correct. A BSDF passing none can still produce an image that looks entirely reasonable, which is precisely why these tests exist rather than relying on inspection.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapters 9 and 14, Reflection Models and Sampling	The complete treatment with code: Fresnel, microfacet distributions, Smith masking, and the sampling routines.
Walter et al. — Microfacet Models for Refraction through Rough Surfaces (2007, free)	The paper that brought GGX into graphics, including the rough transmission derivation of §3.
Heitz — Sampling the GGX Distribution of Visible Normals (2018, free)	The algorithm of §2, with the derivation and complete code. Short paper, large practical payoff.
Heitz — Understanding the Masking-Shadowing Function (2014, free)	Everything about G , including why the height-correlated form is the right default.
Kulla & Conty — Revisiting Physically Based Shading at Imageworks (free)	The energy compensation of §1, from the people who deployed it. The whole SIGGRAPH shading course archive is free and excellent.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement GGX D, the Smith height-correlated G_2 , and Schlick's Fresnel approximation. Verify D integrates correctly over the hemisphere.
- 2 Implement visible-normal sampling and verify it with a χ^2 test. Deliberately omit the $1/(4|\omega_o \cdot h|)$ Jacobian and confirm the test catches what the image does not.
- 3 Compare visible-normal sampling against sampling D directly. Report the variance ratio at normal incidence and at 80° .
- 4 Run the white furnace test across roughness from 0.05 to 1.0 in steps of 0.05 and plot the result. The departure from 1.0 at high roughness is the energy loss of §1 — report how much is lost.
- 5 Implement Kulla–Conty energy compensation and repeat the previous exercise. The curve should flatten to approximately 1.
- 6 Test your BSDF for reciprocity over a thousand random direction pairs and report the maximum relative asymmetry.
- 7 Implement a smooth conductor as a delta BSDF. Confirm NEE skips it and that the resulting mirror has the correct brightness — compare against an analytic Fresnel calculation.
- 8 Implement a rough dielectric with transmission. Render a glass sphere at roughness 0, 0.1, and 0.3 and verify total internal reflection appears where expected.
- 9 Render a sphere grid: roughness across one axis, metallic across the other. This is the standard material validation image and belongs in your portfolio.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the microfacet assumption and explain why roughness is a distribution rather than a blur.
- 2 Write the microfacet BRDF and say what each of D, F, and G contributes and what fails without it.
- 3 Why can only facets whose normal is the half vector contribute?
- 4 Why did GGX displace Beckmann, and what is notable about that outcome?
- 5 What is the height-correlated Smith function, and why is it preferable to the separable form?
- 6 Why does the single-scattering microfacet model lose energy, and how is it recovered in practice?
- 7 What is visible-normal sampling, and why is it better at grazing angles?
- 8 What is the Jacobian in the BSDF sampling PDF, and what happens if it is omitted?
- 9 Why do specular surfaces need a separate path, and what two integrator behaviours depend on the flag?
- 10 Name the three BSDF tests and say what each one catches.